University of Asia Pacific

Department of Computer Science and Engineering

CSE 450: VLSI Lab

LAB REPORT

Experiment Number: 01

Experiment Title: NOR Gate Design & Simulation in Cadence

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1. Experiment Name

NOR Gate Design & Simulation in Cadence

2. Objective

The objective of this experiment is to design a two-input CMOS NOR gate at the transistor level using Cadence Virtuoso Schematic Editor, create a corresponding symbol view for hierarchical use, build a testbench with pulse sources on the inputs, and verify the correct logical behavior of the gate through transient simulation in ADE Assembler/Explorer using the Spectre simulator.

3. Apparatus / Hardware & Software Requirements

- Cadence Virtuoso Schematic Editor XL
- Cadence Virtuoso Symbol Editor
- Cadence ADE Assembler / ADE Explorer
- Spectre Simulator
- 45nm CMOS technology library (g45p1svt PMOS, g45n1svt NMOS devices)

4. Circuit Diagram / Schematic

The NOR gate was implemented at the transistor level using two PMOS transistors in series (pull-up network) and two NMOS transistors in parallel (pull-down network), following standard CMOS NOR gate topology.

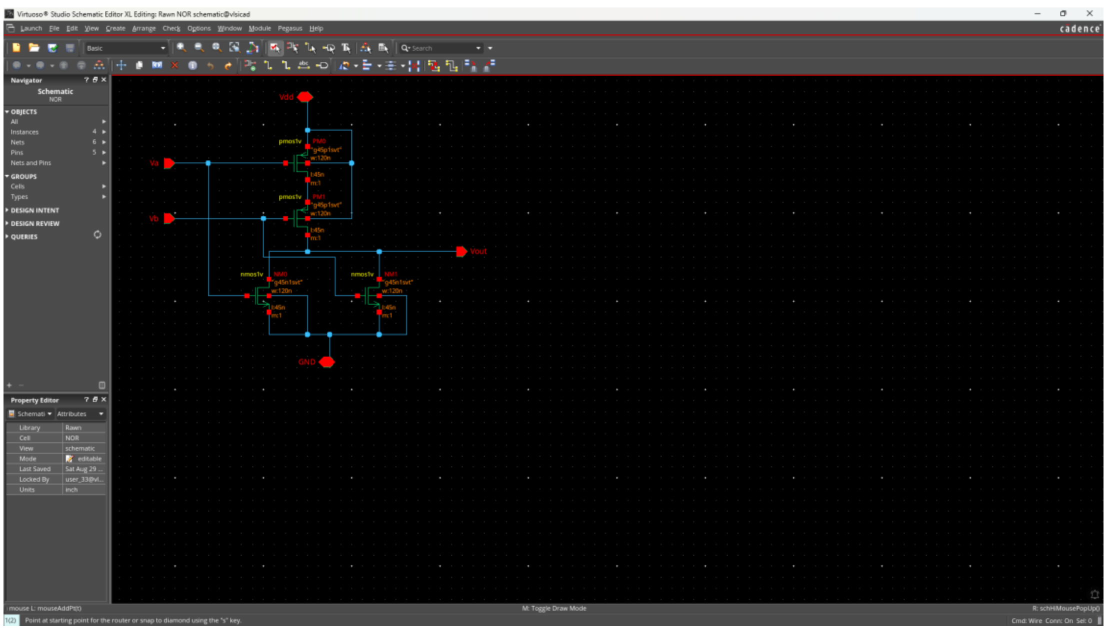

Figure 1: Transistor-level schematic of the CMOS NOR gate

A symbol view was generated from the schematic for use as a black-box instance in the testbench.

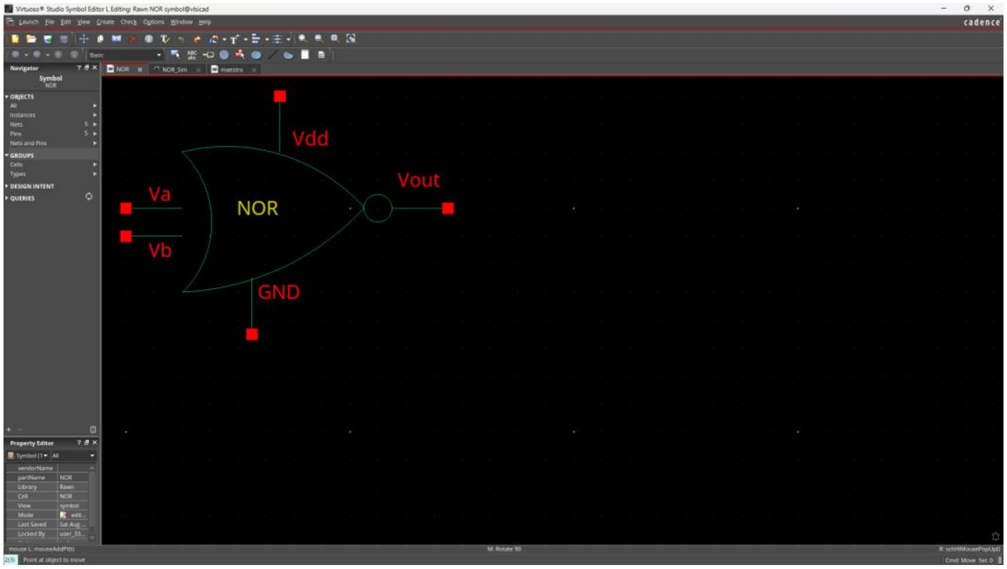

Figure 2: Symbol view of the NOR gate

A testbench was built instantiating the NOR gate symbol with two pulse voltage sources (V1, V2) driving inputs Va and Vb, and a DC supply (Vdd = 1V) powering the gate.

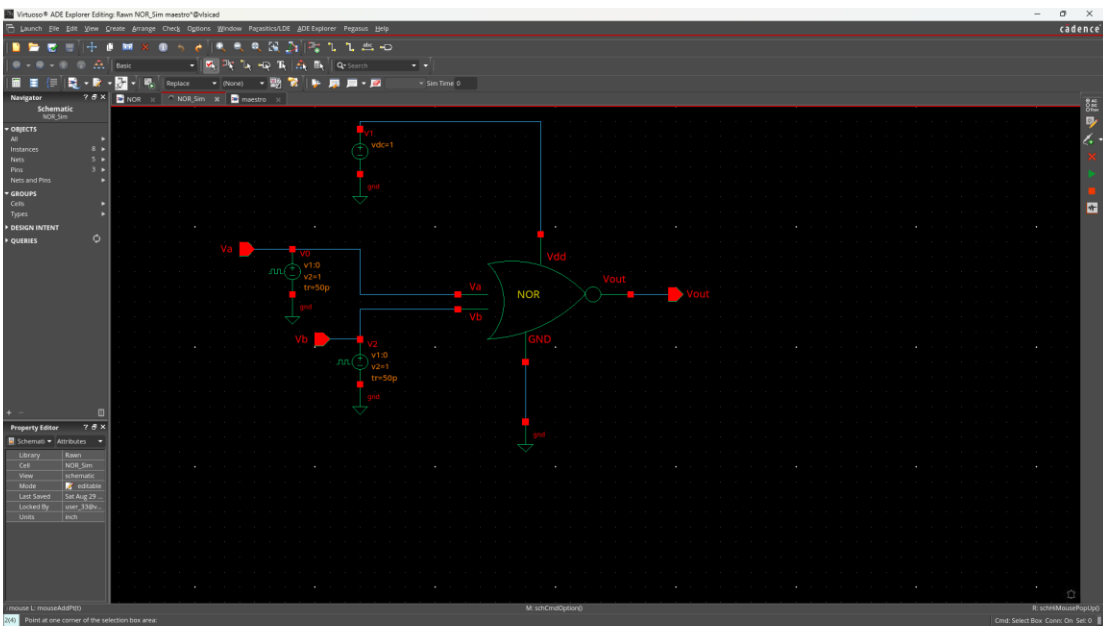

Figure 3: Testbench schematic with pulse sources on Va and Vb

6. Output Analysis / Observations

A transient analysis (0 to 400ps) was run in ADE Assembler with the Spectre simulator. The voltage nodes Va, Vb, and Vout were probed and plotted. The simulation completed with 0 errors and 0 warnings.

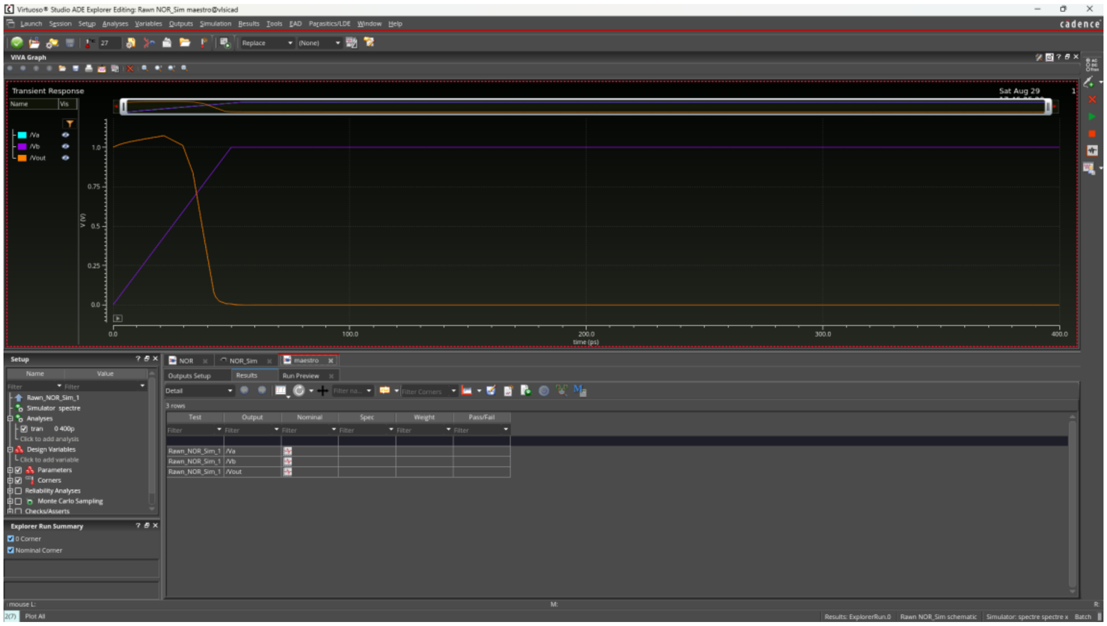

Figure 4: Transient response showing Va, Vb, and Vout

From the waveform, Vout stays high (~1V) while both Va and Vb are low, and drops to 0V as soon as either input transitions high. This matches the expected NOR truth table:

Va	Vb	Vout
0	0	1
0	1	0
1	0	0
1	1	0

7. Result

The designed CMOS NOR gate correctly implements the NOR logic function. The output Vout was observed to be logic HIGH only when both inputs Va and Vb were LOW, and LOW for all other input combinations, matching the expected truth table. The objective of the experiment was successfully achieved.

8. GitHub Repository Commits

GitHub link- https://github.com/rsf-rawnak/CSE450-VLSI-Labs/tree/main/Lab-01-NOR-Gate

9. Conclusion

This experiment provided hands-on experience in designing a CMOS NOR gate at the transistor level in Cadence Virtuoso, creating a reusable symbol view, and setting up a transient simulation testbench using ADE Assembler. It also reinforced the relationship between CMOS pull-up/pull-down network topology and the resulting logic function, and demonstrated how to verify digital gate functionality through waveform analysis.